Uric Acid

Test Details

METHODOLOGY

Uricase

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Uric acid measurements are useful in the diagnosis and treatment of gout, renal failure, and a variety of other disorders including leukemia, psoriasis, starvation, and other wasting conditions. Patients receiving cytotoxic drugs may be monitored with uric acid measurements. Only a minority of individuals with hyperuricemia develop gout. An increased uric acid level does not necessarily translate to a diagnosis of gout.[3] The therapeutic goal for uric acid-lowering therapy is to promote crystal dissolution and prevent crystal formation. This is achieved by maintaining a uric acid level <6 mg/dL.[2]

Elevated uric acid: Elevations of uric acid occur with increased purine synthesis, inherited metabolic disorders, excess dietary purine intake, increased nucleic acid turnover, malignancy, cytotoxic drugs, decreased excretion due to chronic renal failure, and increased renal reabsorption.

Drugs: Drugs causing increased uric acid concentration include diuretics, pyrazinamide, ethambutol, and nicotinic acid.[3]

Endocrine: Hypothyroidism, hypoparathyroidism, hyperparathyroidism, pseudohypoparathyroidism, diabetes insipidus of nephrogenic type, and Addison disease can cause uric acid elevation. Lead poisoning from paint, batteries, and moonshine can cause elevated uric acid.[4] Toxemia of pregnancy, diet, weight loss, fasting, or starvation can elevate uric acid levels.[5]

Decreased uric acid: Drugs bearing a relationship to low serum uric acid levels include aspirin (high doses), x-ray contrast agents, glyceryl guaiacolate, allopurinol corticosteroids, and probenecid.[6] Massive doses of vitamin C increase urine uric acid secretion, lowering serum uric acid.[4] Poor dietary intake of purines and protein can decrease serum uric acid. Diabetes, Fanconi syndrome, Wilson's disease, cystinosis, galactosemia, hypophosphatemia, heavy metal poisoning, malignant neoplasms, hypereosinophilic syndrome, and Xanthinuria (deficiency of xanthine oxidase) can lower serum uric acid.[4,6,7] Hypouricemia is reported with acute intermittent porphyria and severe liver disease (especially obstructive biliary disease).[8] Isolated defects in the tubular transport of uric acid have been associated with increased renal clearance of urate, hypouricemia, hypercalciuria, and decreased bone density.[9]

Custom Additional Information

Drug effects have been summarized.[10]

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

Separate serum or plasma from cells within 45 minutes of collection.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

• Male:[1]

– 0 to 30 days: 3.9−7.8 mg/dL

– 1 to 6 months: 1.9−8.1 mg/dL

– 7 to 11 months: 2.0−6.5 mg/dL

– 1 to 11 years: 2.2–5.5 mg/dL

– 12 years: 2.9–7.0 mg/dL

– 13 to 17 years: 3.9–7.7 mg/dL

– 18 years and older: 3.8–8.4 mg/dL

• Female:

– 0 to 30 days: 2.7–6.5 mg/dL

– 1 to 6 months: 2.0–6.6 mg/dL

– 7 to 11 months: 2.1–5.7 mg/dL

– 1 to 5 years: 2.0–5.0 mg/dL

– 6 to 11 years: 2.4–5.6 mg/dL

– 12 to 17 years: 2.9–6.1 mg/dL

– 18 to 50 years: 2.6–6.2 mg/dL

– 51 to 70 years: 3.0–7.2 mg/dL

– 71 years and older: 3.1–7.9 mg/dL

Therapeutic target for gout patients: <6.0[2]

Storage Instructions

Maintain specimen at room temperature.[1]

Causes for Rejection

Improper labeling

Footnotes

1. LabCorp internal studies

2. Zhang W, Doherty M, Bardin T, Suárez DH, Aristimuno GG. EULAR evidence based recommendations for gout. Part II: Management. Report of a task force of the EULAR Standing Committee for International Clinical Studies Including Therapeutics (ESCISIT). *Ann Rheum Dis.* 2006 Oct; 65(10):1301-1311. PubMed 16707532 (http://www.ncbi.nlm.nih.gov/pubmed/16707532)

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5. Lin HY, Rocker LL, McQuillan MA, Schmaltz S, Palella TD, Fox IH. Cyclosporine-induced hyperuricemia and gout. *N Engl J Med.* 1989 Aug 3; 321(5):287-292. PubMed 2664517 (http://www.ncbi.nlm.nih.gov/pubmed/2664517)

6. Lugassy G, Michaeli J. Hypouricemia in the hypereosinophilic syndrome. Response to treatment. *JAMA.* 1983 Aug 19; 250(7):937-938. PubMed 6864977 (http://www.ncbi.nlm.nih.gov/pubmed/6864977)

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8. Ramsdell CM, Kelley WN. The clinical significance of hypouricemia. *Ann Intern Med.* 1973 Feb; 78(2):239-242. PubMed 4683752 (http://www.ncbi.nlm.nih.gov/pubmed/4683752)

9. Steele TH. Hypouricemia. *N Engl J Med.* 1979 Sep 6; 301(10):549-550. PubMed 460310 (http://www.ncbi.nlm.nih.gov/pubmed/460310)

10. Zhiri A, Jouanel P. Urates. In: Siest G, Galteau MM, eds. *Drug Effects on Laboratory Test Results. Analytical Interferences and Pharmacological Effects.* Littleton, Mass: PSG Publishing Co;1988:423-438. PubMed 3606290 (http://www.ncbi.nlm.nih.gov/pubmed/3606290)